

Day 4: Other Protocols in ActionTheory (30 min):Introduction to FTP and SNMP.Practical (30 min):Set up an FTP server and transfer files between devices.Task: Create a table comparing FTP and HTTP.

Task 1.

Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Administration interface - FileZilla Server 1.12.6

Server Window Help			
Date/Time	Info	Type	Message
05-08-2026 22:01:46	Admin UI	Status	Successfully connected to server 127.0.0.1:14148. Server's version is 1.12.6, built for x86_64-w64-mingw32.
05-08-2026 22:03:22	FTP Session 3 192.168.56.1	Response	220-FileZilla Server 1.12.6
05-08-2026 22:03:22	FTP Session 3 192.168.56.1	Response	220 Please visit https://filezilla-project.org/
05-08-2026 22:03:22	FTP Session 3 192.168.56.1	Comma...	AUTH TLS
05-08-2026 22:03:22	FTP Session 3 192.168.56.1	Response	234 Using authentication type TLS.
05-08-2026 22:03:39	FTP Session 3 192.168.56.1	Comma...	USER student
05-08-2026 22:03:39	FTP Session 3 192.168.56.1	Response	331 Please, specify the password.
05-08-2026 22:03:39	FTP Session 3 192.168.56.1	Comma...	PASS ****
05-08-2026 22:03:39	FTP Session 3 192.168.56.1	Response	230 Login successful.
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Comma...	FEAT
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	211-Features:
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	211 End
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Comma...	PWD
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	257 "/" is current directory.
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Comma...	TYPE I
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	200 Type set to I
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Comma...	PASV
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	227 Entering Passive Mode (192,168,56,1,253,235)
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Comma...	MLSD
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	150 Starting data transfer.
05-08-2026 22:03:39	FTP Session 3 192.168.56....	Response	226 Operation successful
05-08-2026 22:03:59	FTP Session 3 192.168.56....	Comma...	CDUP
05-08-2026 22:03:59	FTP Session 3 192.168.56....	Response	250 CWD command successful
05-08-2026 22:03:59	FTP Session 3 192.168.56....	Comma...	PWD
05-08-2026 22:03:59	FTP Session 3 192.168.56....	Response	257 "/" is current directory.

Date/Time	Session ID	Protocol	Host	Username	Transfer
05-08-2026 22:03:22	3	FTPS	192.168....	student	Idle

student@192.168.56.1 - FileZilla

File Edit View Transfer Server Bookmarks Help

Host: 192.168.56.1 Username: student Password: Port: Quickconnect

Status: Retrieving directory listing...
Status: Directory listing of "/" successful
Status: Retrieving directory listing of "/"...
Status: Directory listing of "/" successful

Local site: C:\Users\sketan\

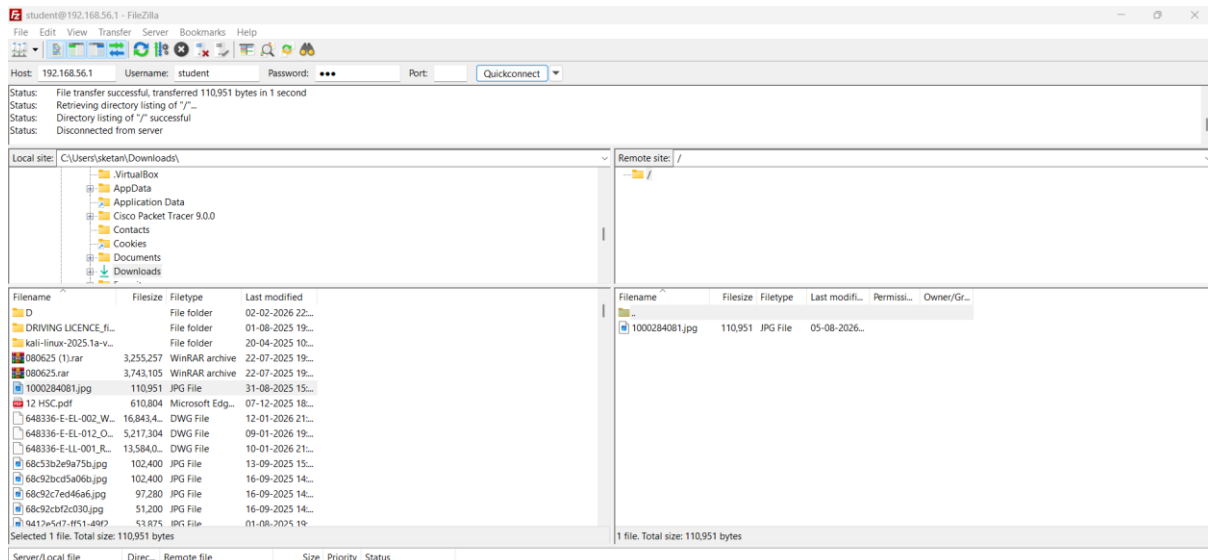
Remote site: /

Filename	Filesize	Filetype	Last modified	Filename	Filesize	Filetype	Last modified	Permissi...	Owner/Gr...
.				.					
VirtualBox		File folder	05-08-2026 18...						
AppData		File folder	20-12-2024 04...						
Application Data		File folder	05-08-2026 22...						
Cisco Packet Tracer ...		File folder	04-08-2026 19...						
Contacts		File folder	20-12-2024 04...						
Cookies		File folder	13-04-2025 16...						
Documents		File folder	13-04-2025 19...						
Downloads		File folder	05-08-2026 21...						

Empty directory listing

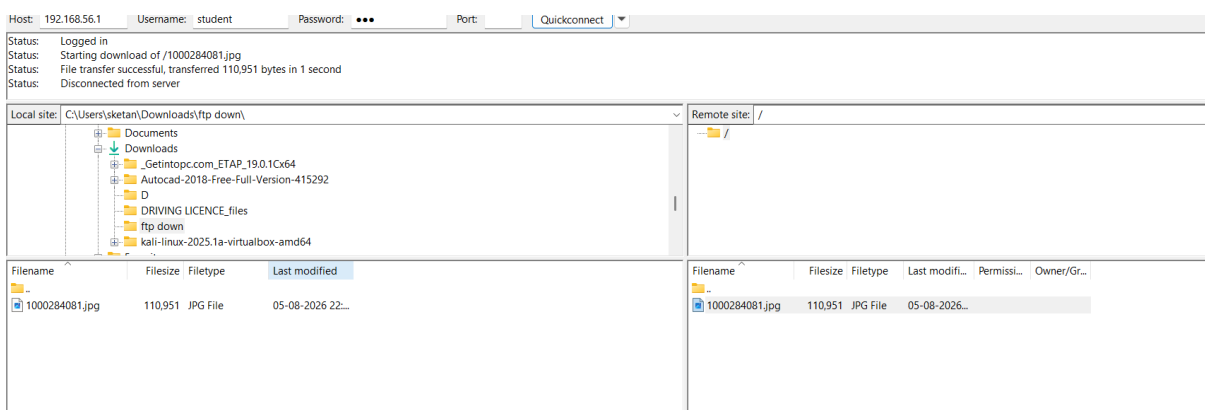
Task 2.

Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.



Task 3.

Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.



Task 4.

Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Hint: Think about how network admins monitor device health, bandwidth, or uptime.

Example	Use
ISP Networks	ISPs monitor routers, switches, bandwidth usage, interface status, and uptime to quickly detect outages and congestion.
Campus Wi-Fi	Universities use SNMP to monitor wireless access points, client connections, network traffic, and device health for reliable campus internet.
Large Websites and Data Centers	Companies running large web services use SNMP to monitor servers, switches, firewalls, CPU usage, memory utilization, and network availability to maintain uptime.

Task 5.

Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Feature	FTP	HTTP
Port Number	21 (control), 20 (data in active mode)	80 (HTTP), 443 (HTTPS)
Security	Plain FTP is unencrypted; secure variants include FTPS and SFTP (SFTP is a different protocol)	HTTPS encrypts communication using TLS
Data Transfer Method	Dedicated protocol for uploading and downloading files between client and server	Transfers web pages and other web resources using a request-response model
Authentication	Typically requires a username and password (anonymous access is also possible)	Public websites often require no login; authenticated web applications use usernames, passwords, tokens, etc.
Common Use Cases	File sharing, backups, website file uploads, internal file transfers	Browsing websites, web applications, APIs, streaming content, online services